

DIPOLE TYPE BY LENGTH

STEP 0 — always start here $\lambda = \text{wavelength (m)}, l = \text{antenna rod length (m)}, c = 3 \times 10^8 \text{ m/s}$

$\lambda = \frac{c}{f} \text{ (m)}$	ratio $\frac{l}{\lambda} \rightarrow$	pick row below
l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D = 1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave shortcut, $D = 1.64$, $R_{\text{rad}} = 73 \Omega$
$= 1$	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}} = 199 \Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$
$\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$, $\vec{H}_\phi = \vec{E}_\theta / \eta_0$
Power density $l/\lambda < 1/50$; far field
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2}{32\pi^2 R^2} \sin^2 \theta = S_{\text{max}} \sin^2 \theta \text{ (W/m}^2\text{)}$
l rod length (m); I_0 peak current (A); R obs. dist. (m); θ from dipole axis; $k = 2\pi/\lambda$ (rad/m); $\eta_0 = 120\pi \approx 377 \Omega$.
Max @ broadside $\theta_{\text{max}} = \pi/2$; D exact $D = 1.5$ (1.76 dB)
Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2 \text{ (W)}$

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R any l
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$
At $\theta = 0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow S = 0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta = 0.001$ rad $\rightarrow$ tiny $\rightarrow 0$.
Normalized pattern dimensionless $F(\theta) = S(\theta)/S_{\text{max}}$
Quarter-wave $l = \lambda/4$ $\pi l/\lambda = \pi/4$, $\cos(\pi/4) = \sqrt{2}/2$
$\frac{S(\theta)}{S_0} = \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$, $S_0 = \frac{15 I_0^2}{\pi R^2}$
$S_{\text{max}} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$
$D = 1.64$, $R_{\text{rad}} = 36.5 \Omega$, $\theta_{\text{max}} = \pi/2$.
Half-wave $l = \lambda/2$ $\pi l/\lambda = \pi/2$, shortcut form
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}$
$D = 1.64$, $R_{\text{rad}} = 73 \Omega$, $\theta_{\text{max}} = \pi/2$.

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

Conversion multiply by $\pi/180$ or $180/\pi$
$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$ $\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$
$^\circ$ rad $\sin \theta$ $\cos \theta$ $\sin^2 \theta$ $\cos^2 \theta$
0 0 0 1 0 1
30 $\pi/6$ 1/2 $\sqrt{3}/2$ 1/4 3/4
45 $\pi/4$ $\sqrt{2}/2$ $\sqrt{2}/2$ 1/2 1/2
60 $\pi/3$ $\sqrt{3}/2$ 1/2 3/4 1/4
90 $\pi/2$ 1 0 1 0
180 π 0 -1 0 1

Antenna formulas usually take θ in rad. RAD mode on calc.

DIPOLE

l — antenna length (m)
 $\lambda = c/f$ (m); $c = 3 \times 10^8 \text{ m/s}$
 I_0 — peak current (A)
 $k = 2\pi/\lambda$ (rad/m)
 R — obs. dist. (m); θ from axis
 $\eta_0 = 120\pi \approx 377 \Omega$
 $S(R, \theta)$ — pwr density (W/m²)

BEAM

$F(\theta) = S/S_{\text{max}}$ (unitless)
 β — HPBW (rad or $^\circ$)
 Ω_p — pattern solid angle (sr)
 $D = 4\pi/\Omega_p$ — directivity
 ξ — rad. efficiency (0–1)
 $G = \xi D$ — gain (unitless)
 $D_{\text{dB}} = 10 \log_{10} D$

FRIIS / dB

P_t, P_r — TX/RX power (W)
 G_t, G_r — linear gains
 $S_r = G_t P_t / (4\pi R^2)$ (W/m²)
 $P_r = G_t G_r (\lambda/4\pi R)^2 P_t$
 $G = 10^{G_{\text{dB}}/10}$
 $3 \text{ dB} = \times 2, 10 \text{ dB} = \times 10$
 $P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

A_p — physical area (m²)
 $A_e = \lambda^2 D / (4\pi)$ (m²)
 $A_e \approx A_p$ for aperture
 $D \approx 4\pi A_p / \lambda^2$ (unitless)
 $\beta \approx 0.88\lambda/\ell$ (rect/sq.)
 $\beta \approx 1.02\lambda/d$ (circular)
 $2A_p \Rightarrow 2D, \beta/\sqrt{2}$

ARRAY

N — number of elements
 d — spacing (m)
 $\gamma = kd \cos \theta$
 $F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$
 $F_a^{\text{max}} = N^2$ at $\theta = \pi/2$
 $F_a^{\text{norm}} = F_a / N^2$
 $\beta \approx 0.88\lambda/(Nd)$ (broadside)

BEAM PARAMETERS — β, Ω_p, D

Step 1 — Normalize always start here
$F(\theta) = S(\theta)/S_{\text{max}}$ (unitless)
Step 2 — HPBW β half-power beamwidth, full angular width at -3 dB
Set $F(\theta) = 0.5$, solve for $\theta \rightarrow \theta_{1/2}$
$\beta = 2\theta_{1/2}$ (symmetric beams, rad or $^\circ$)
Gaussian:
$F(\theta_{1/2}) = e^{-a\theta_{1/2}^2} = 0.5 \Rightarrow \theta_{1/2} = \sqrt{\ln 2/a}$
Step 3 — Pattern solid angle Ω_p units: sr
$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta d\theta d\phi$
No ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta d\theta$.
TI-36X Pro: [2nd] [d/dx] for $\int_0^\pi$, set RAD mode, multiply result by 2π .
Narrow-beam Gaussian shortcut ($\sin \theta \approx \theta, \rightarrow \infty$):
$\Omega_p \approx \frac{\pi}{a} = \frac{\pi \theta_{1/2}^2}{\ln 2}$ ($F = e^{-a\theta^2}$)

Step 4 — Directivity D
$D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$
Alt. with two orthogonal HPBW's: $D \approx 4\pi/(\beta_{xz}\beta_{yz})$.
Step 5 — Gain (if asked) $\xi = \text{rad. eff.}, 0 < \xi \leq 1$
$G = \xi D$ $G_{\text{dB}} = 10 \log_{10} G$
Lossless / ideal antenna $\Rightarrow G = D$.

COMMON DIPOLE PARAMETERS

Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
$\lambda/4$	$\lambda/4$ (gnd)	1.64	2.15	36.6
monopole				
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{\text{rad}} = 36.6 I_0^2 W$. $\lambda/4$ monopole (above ground): same patt. as half-wave but $P_{\text{rad}}, R_{\text{rad}}$ halved.
 Lossless antenna ($\xi = 1$) $\Rightarrow G = D$. So in Friis, use $G_t, G_r =$ these D values (e.g. half-wave $G = 1.64$).

EFFECTIVE AREA A_e

Definition antenna's RX cross section (m²)
$A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}$
Physical cross section wire dipole, diameter d
$A_p = l d = \frac{\lambda}{2} d$ (half-wave)
Inverse: D from A_e
$D = \frac{4\pi A_e}{\lambda^2}$ (unitless)
Hertzian special case $D = 1.5$
$A_e^{\text{Hertz}} = \frac{3\lambda^2}{8\pi} \approx 0.119\lambda^2 \text{ (m}^2\text{)}$
Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim \lambda$.

LINEAR ANTENNA ARRAY

Setup N identical elements along axis, spacing d
elem pos $z_i = id, i = 0, \dots, N-1; A_i = a_i e^{j\psi_i}$
Equal phase ($\psi_i = 0$) $\Rightarrow$ broadside beam ($\theta_{\text{max}} = \pi/2$).
Progressive phase $\psi_i = i\alpha \Rightarrow$ steered beam at $\cos \theta_{\text{max}} = -\alpha/(kd)$.
Pattern multiplication S_e = single element pattern
$S(R, \theta, \phi) = S_e(R, \theta, \phi) F_a(\theta)$
Array factor (general) $k = 2\pi/\lambda$
$F_a(\theta) = \left \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta} \right ^2$
Uniform $A_i = 1$, equal phase closed form
$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}, \gamma = kd \cos \theta$
Peak: $F_a^{\text{max}} = N^2$ at $\gamma = 0$ (broadside, $\theta = \pi/2$). Normal-ize: $F_a^{\text{norm}} = F_a/N^2$.
HPBW (broadside, uniform) use Nd, not $(N-1)d$
$\beta \approx 0.88 \frac{\lambda}{Nd}$ (rad) $= 50.8^\circ \cdot \frac{\lambda}{Nd}$
Grating lobes appear if $d > \lambda$ (broadside). Avoid.

FRIIS TRANSMISSION FORMULA

Pwr density at RX, then RX pwr G_t, G_r linear gains; $\lambda = c/f$

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{)}; P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \text{ (W)}$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e recu. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) when patterns NOT peak-aligned

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert dB gains to linear: $G = 10^{G_{\text{dB}}/10}$.

3. If TX is "half-wave dipole" use $G_t = 1.64$. 4. Plug into Friis.

Use linear G , not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity: G gain, D directivity, P_r/P_t , SNR, etc.

$$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}} \quad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

e.g. $D_{\text{dB}} = 10 \log_{10} D$; $G_{\text{dB}} = 10 \log_{10} G$;
 $SNR_{\text{dB}} = 10 \log_{10} (P_r/P_N)$.

	dB linear	dB linear
0	1	10
3	2	13
6	4	20
7	5	30
10	10	1000
-3	0.5	-10
-0.1		0.1

Memorize: $3 \text{ dB} \Leftrightarrow \times 2, 10 \text{ dB} \Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use $20 \log$ not $10 \log$ (power $\propto E^2$).

NOISE & SNR

Noise power thermal noise into matched receiver
$P_N = k_B T_{\text{sys}} B$ (W)
$k_B = 1.38 \times 10^{-23} \text{ J/K}$; T_{sys} system noise temp (K); B receiver bandwidth (Hz).
Signal-to-noise ratio
$SNR = \frac{P_{\text{rec}}}{P_N}$ (unitless) $SNR_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$
Typical comm. link needs $SNR > 10 \text{ dB}$ for usable signal.

APERTURE ANTENNAS (horn, dish)

Directivity (any shape, uniform illum.)
$D \approx \frac{4\pi A_p}{\lambda^2}$ (unitless) $\Rightarrow A_e \approx A_p$
Directivity from beamwidths any aperture
$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}}$ (use rad); circular: $D \approx 4\pi/\beta^2$

HPBW and D by aperture shape uniform illum.; β in rad

Shape	A_p	HPBW (rad)	D (unitless)
Rect. $\ell_x \times \ell_y$	$\ell_x \ell_y$	$\beta_x \approx 0.88\lambda/\ell_x$ $\beta_y \approx 0.88\lambda/\ell_y$	$D \approx 4\pi/(\beta_x \beta_y)$ $= 4\pi \ell_x \ell_y / \lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88\lambda/\ell$	$D \approx \frac{4\pi}{\beta^2} = \frac{4\pi \ell^2}{\lambda^2}$
Circular (diam. d)	$\pi d^2/4$	$\beta \approx 1.02\lambda/d$	$D \approx \frac{4\pi}{\beta^2} = \frac{\pi^2 d^2}{\lambda^2}$
Scaling rules $D \propto A_p/\lambda^2; \beta \propto \lambda/\sqrt{A_p}$			
Change	New D	New β	
Double ($A_p \rightarrow 2A_p$)	area	$D_{\text{new}} = 2D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}}/\sqrt{2}$
Double freq ($\lambda \rightarrow \lambda/2$)	$D_{\text{new}} = 4D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}}/2$	